

PROJECT ABSTRACT

Real-Time Air Quality and Gas Leak Monitoring System

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With cities growing faster than ever, keeping the air we breathe safe from pollution and gas leaks is a huge priority. For our project, we are building a simple, affordable smart monitor that checks air quality in real time.

Using an MQ-135 gas sensor, our device constantly sniffs the air for harmful gases like carbon dioxide, ammonia, and smoke. The microcontroller acts as the brain of the operation, taking the raw sensor data and turning it into an easy-to-understand air quality score. To make sure everyone stays safe, the system gives instant feedback: a small screen shows the live readings, while a set of traffic-light LEDs (Green, Yellow, Red) shows the current safety level at a glance. If things get dangerous, a buzzer sounds a loud alarm to warn anyone nearby. This automated setup offers a scalable, user-friendly safety solution for homes, schools, and workplaces.

1. Parts We Need

Here is the list of parts we need to bring this project to life. We designed this to be easy to build using standard, beginner-friendly components.

Component	Quantity	What it does
Microcontroller Board	1	An Arduino Uno R3, Nano, or ESP32. This is the brain of the project that reads the sensor data and controls the screen, lights, and buzzer.
MQ-135 Gas Sensor	1	Our electronic "nose." It detects a wide range of gases including smoke, alcohol, and CO2.
16x2 I2C LCD Display	1	A small screen that shows the live air quality numbers so we can easily read them.
Active Piezo Buzzer	1	Acts as our alarm system. It makes a loud beeping noise if the air gets toxic.
LED Indicators	3	Green (Safe), Yellow (Warning), and Red (Danger) lights to show the status at a glance.
Resistors (220Ω - 330Ω)	3	These protect our LEDs from getting too much power and burning out.
Breadboard & Jumper Wires	1 Set	The board and wires let us plug everything together easily without having to solder anything permanently.
Power Supply	1	A standard 9V battery or a USB power bank to keep the project running without being plugged into a computer.

2. Wiring It Together

Connecting everything properly is key to getting accurate readings. Here is how we will wire the components to the microcontroller.

Component Pin	Where it connects on the board
MQ-135: Power (VCC) & Ground (GND)	5V & GND
MQ-135: Analog Out (A0)	Analog Pin A0
LCD Screen: Power (VCC) & Ground (GND)	5V & GND
LCD Screen: SDA & SCL (Data lines)	Analog Pins A4 & A5
Green LED: Long leg (+)	Digital Pin 2 (with a resistor)
Yellow LED: Long leg (+)	Digital Pin 3 (with a resistor)
Red LED: Long leg (+)	Digital Pin 4 (with a resistor)
Buzzer: Positive pin (+)	Digital Pin 5
All LEDs & Buzzer: Short legs (-)	Ground (GND)

3. How the Code Works

The code running our system is built around a simple, continuous loop that keeps a constant eye on the environment.

- Starting Up:** When we turn the system on, the code sets up the screen and gives the gas sensor a moment to warm up so it can give us accurate readings.
- Reading the Air:** The microcontroller constantly asks the sensor, "What's the air like right now?" It takes that raw voltage reading and converts it into a simple number we can use.
- Making Decisions (The Smart Part):**
 - **Safe Zone (Reading < 300):** The air is clean. We turn on the Green LED, make sure the buzzer is quiet, and the screen happily reports "Air Quality: GOOD".
 - **Warning Zone (Reading 300 - 600):** The air is getting a bit polluted or smoky. The Yellow LED turns on instead, and the screen warns "Air Quality: POOR".
 - **Danger Zone (Reading > 600):** Gas levels are critical. The Red LED lights up, the buzzer sounds a loud alarm, and the screen flashes "ALERT: GAS DETECTED".
- Updating the Screen:** Finally, the system updates the live numbers on the screen. We add a tiny half-second pause so the screen doesn't flicker and the numbers are easy to read.